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2

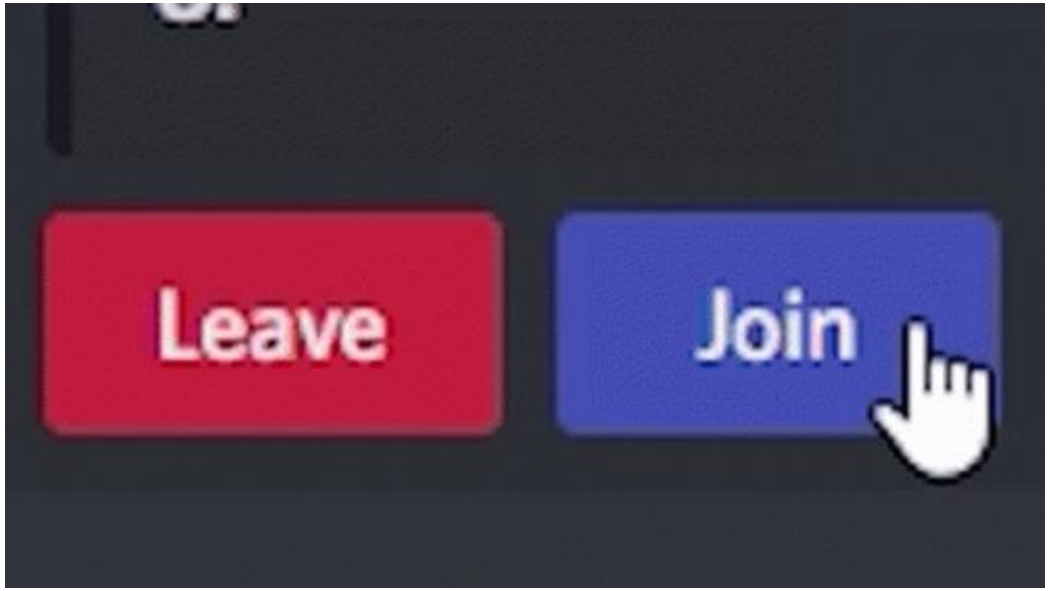
Data Preprocessing

Rules

1. 75% attendance mandatory as per university rules
2. No late entries permitted
3. Laptop screens down is the default state unless specified otherwise
4. Using phones, smartwatches, earphones or any other electronic devices is NOT permitted in class unless instructed otherwise.
5. Students must occupy the front seats first, class would not start if the front seats are vacant
6. When you have a doubt in class, raise your hand, stand up when pointed by the instructor and then ask!
7. Be respectful to everyone.

* This is not an exhaustive list of rules…

Join the Lecture on NS Portal :)



Recap

- Supervised Learning: Model learns from labelled data

- Unsupervised Learning: Model learns from unlabelled data.

- Reinforcement learning: Model learns from continuous feedback. E.g. ChatGPT

Recap

- Classification: Assigning input to discrete category or label. E.g. Cats Vs Dogs, Covid Positive Vs Covid Negative

- Regression: Predicting continuous variable. E.g. height, temperature, marks

Agenda

- Data Ingestion
- Data Inspection
- Statistical Measures
- Missing Data Imputation
- Outliers
- Encoding

?

Benefits of Data Cleaning

- Improved Accuracy
- Better performance
- Increased reliability

Data Cleaning Pipeline

Raw Data → Ingestion → Cleaning → Encoding → Model

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Data Ingestion

LOADING RAW DATA

Pandas

- Easy data loading
- Powerful data structures
- Excellent handling of missing data
- Fast and expressive data cleaning
- Flexible data transformation

Data Ingestion

Process of extracting data from source and loading into storage medium or processing engine. Common sources

- Flat Files: CSV, Excel, Parquet
- Databases: SQL, MongoDB
- Web: APIs, HTML Scraping

Data Inspection

DataFrame(df) is a 2 dimensional data structure with inbuilt features

- df.head(n)
- df.tail(n)
- df.sample(n)

Data Inspection

- df.head(n): Shows the first n rows.
- df.tail(n): Shows the last n rows.
- df.sample(n): Returns n random rows.

Data Inspection

- df.info()

```
RangeIndex: 1000 entries, 0 to 999
Data columns (total 3 columns):
# Column Non-Null Count Dtype
--- ------ -------------- -----
0 Age 950 non-null float64
1 City 1000 non-null object
2 Salary 980 non-null float64
```

DIY

For the given data

[2, 4, 8, 6, 10, 12, 14]

Calculate

- Mean: Sum of observations/number of observations
- Median: (n+1)/2 (for odd number of values)
- Standard deviation: $$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{N}}$$

Source: https://byjus.com/standard-deviation-formula/

DIY

For the given data

[2, 4, 8, 6, 10, 12, 14]

Calculate

- Mean: 8
- Median: 8
- Standard deviation: 4

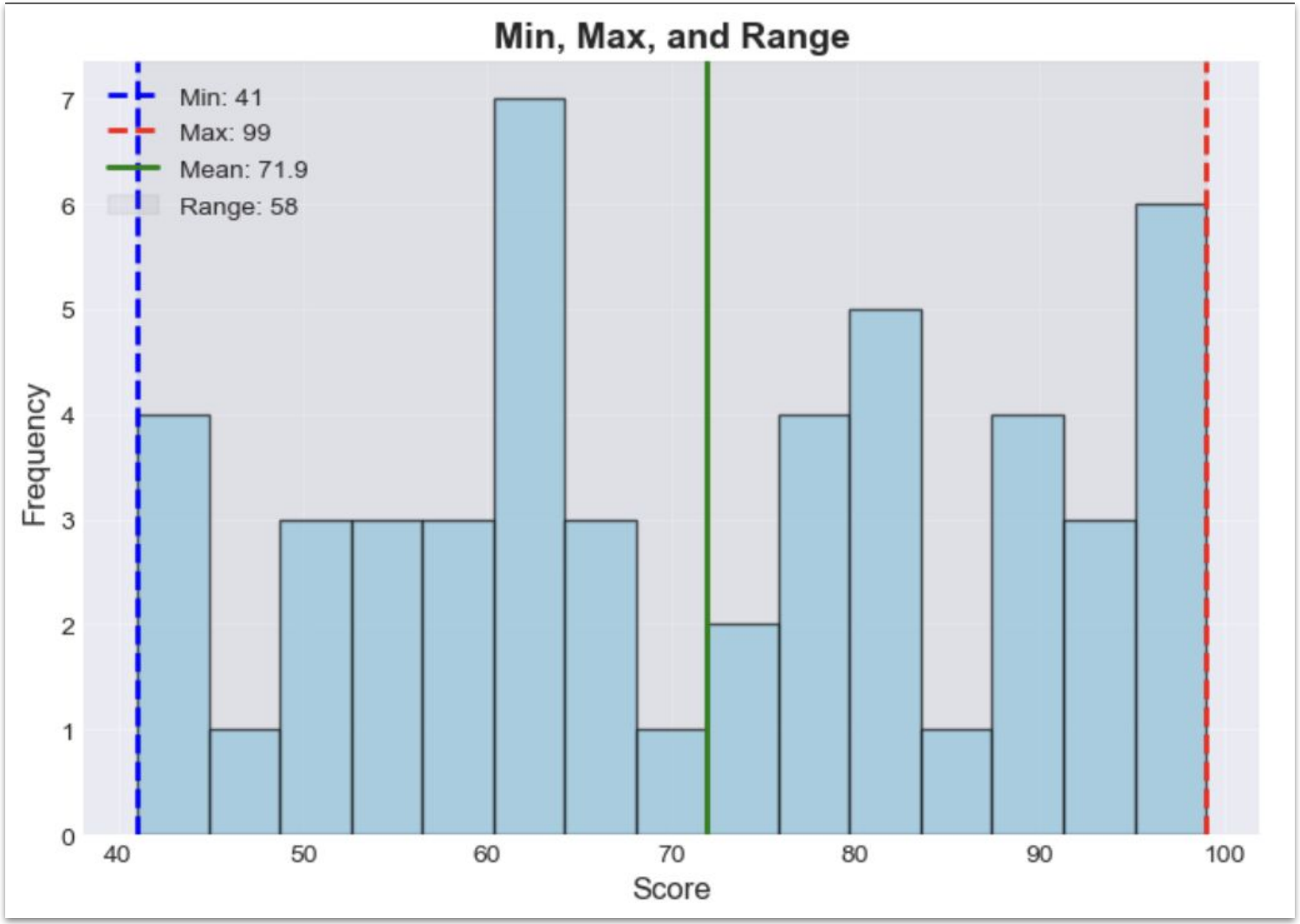
Data Summary

- df.describe()

```
>>> df.describe(exclude=[object])
       categorical  numeric
count            3      3.0
unique           3      NaN
top              f      NaN
freq             1      NaN
mean           NaN      2.0
std            NaN      1.0
min            NaN      1.0
25%            NaN      1.5
50%            NaN      2.0
75%            NaN      2.5
max            NaN      3.0
```

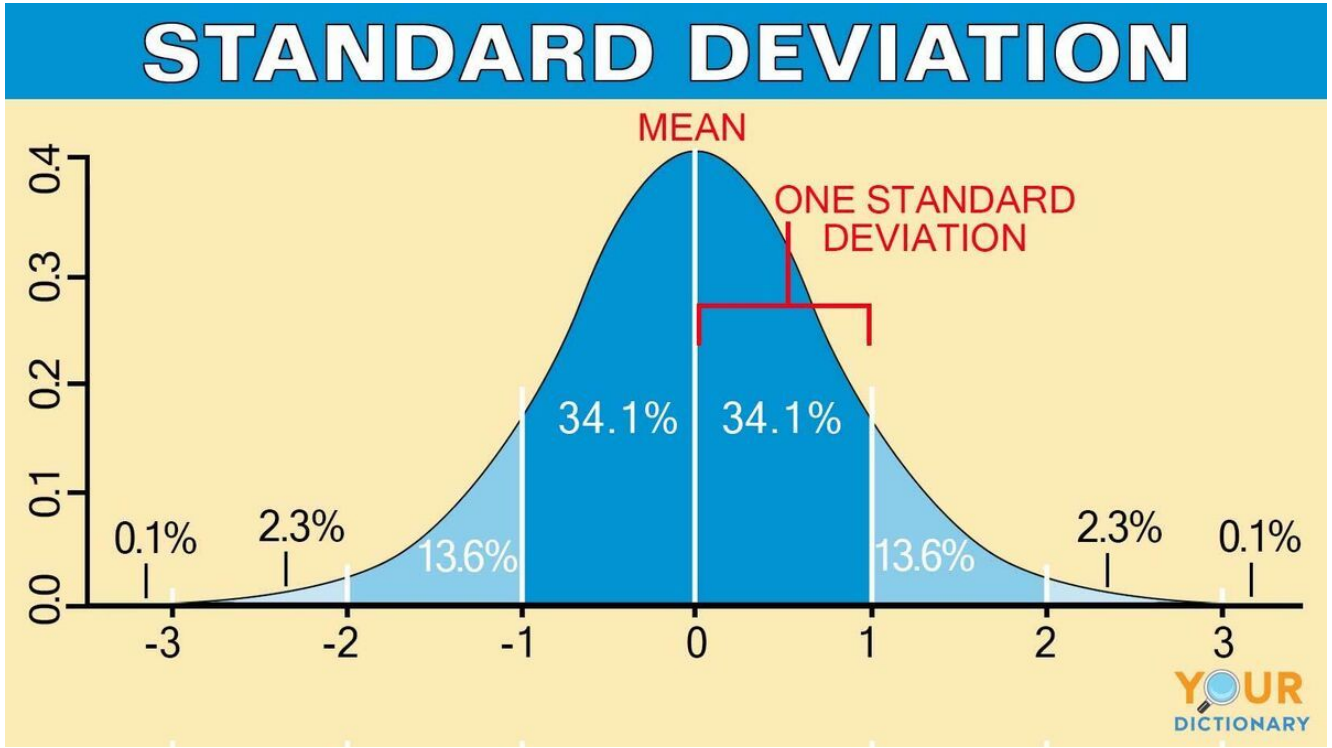
Statistical Summary

- Min, Max and range



Statistical Summary

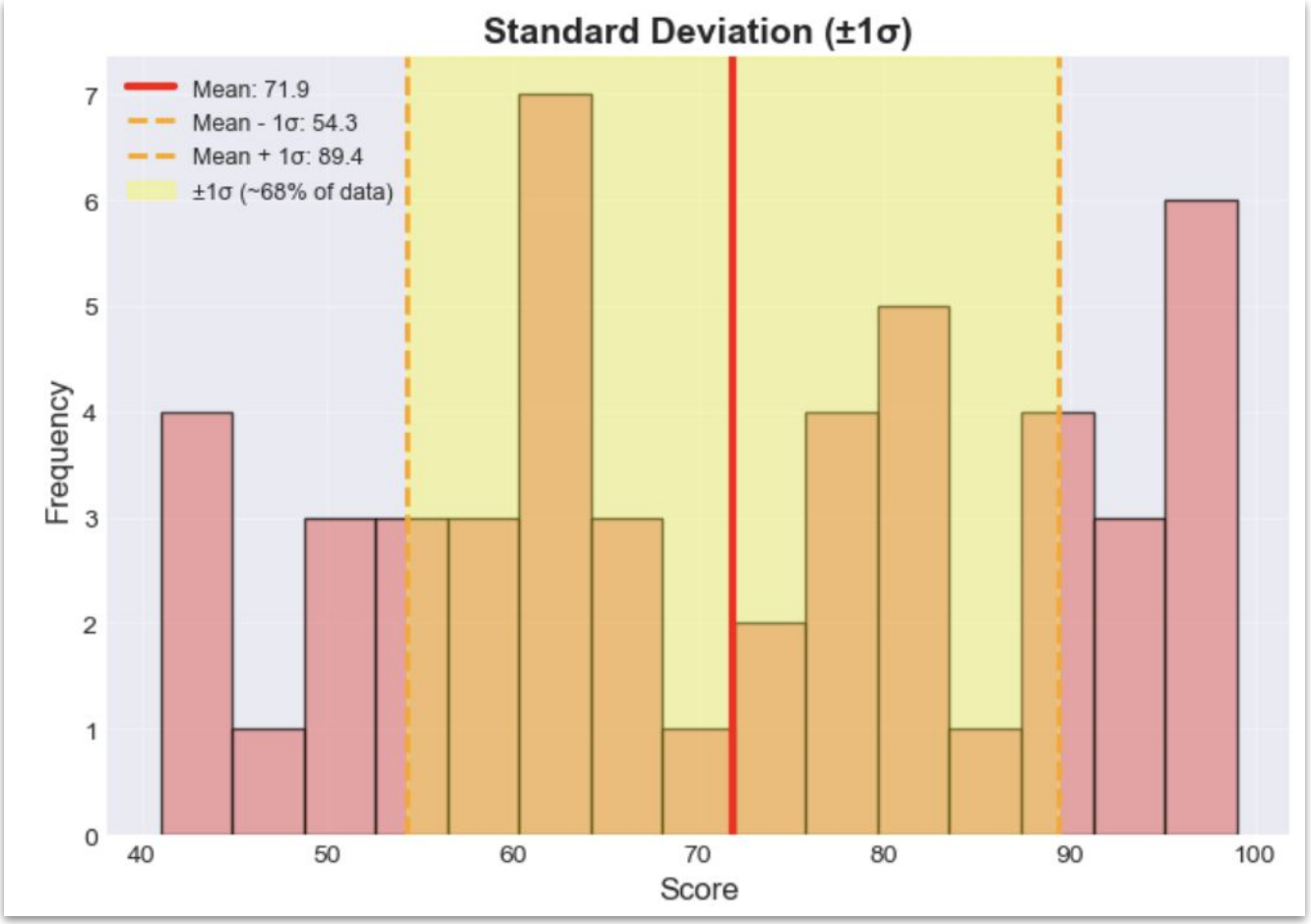
- Standard deviation



Source: https://www.yourdictionary.com/articles/standard-deviation-examples

Statistical Summary

- Standard deviation



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Missing Data

IMPUTATION STRATEGIES

Missing Values

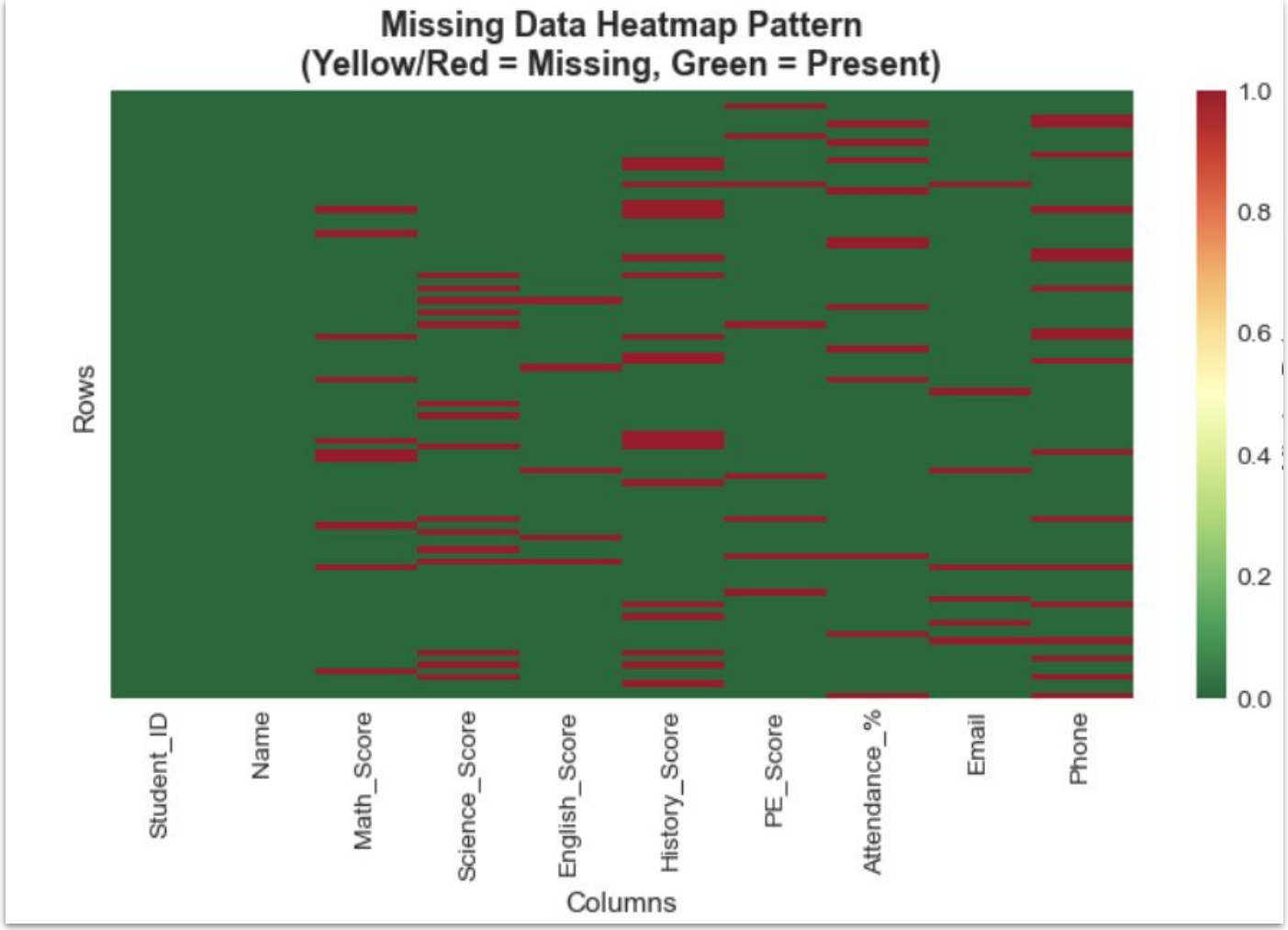
```
   Student_ID        Name  Math_Score  Science_Score  English_Score  \
0           1   Student_1        88.0           63.0           76.0
1           2   Student_2        78.0           78.0           61.0
2           3   Student_3        64.0           55.0           61.0
3           4   Student_4        92.0           62.0           87.0
4           5   Student_5        57.0           78.0           82.0
5           6   Student_6        70.0           65.0           91.0
6           7   Student_7        88.0           71.0           92.0
7           8   Student_8        68.0           62.0           60.0
8           9   Student_9        72.0           89.0           78.0
9          10  Student_10        60.0           89.0           61.0

   History_Score  PE_Score  Attendance_%                 Email     Phone
0           65.0      73.0          78.0   student1@school.edu  555-6588
1           74.0      75.0          90.0   student2@school.edu  555-8651
2           77.0       NaN          98.0   student3@school.edu  555-4636
3           72.0      89.0          90.0   student4@school.edu  555-2167
4           91.0      99.0          97.0   student5@school.edu       NaN
5           77.0      84.0           NaN   student6@school.edu       NaN
6           49.0      72.0          91.0   student7@school.edu  555-8860
7           92.0       NaN          94.0   student8@school.edu  555-6161
8           63.0      77.0           NaN   student9@school.edu  555-6305
```

- Can you tell me the columns that have missing values?

Missing Data Heatmap

```
import seaborn as sns
sns.heatmap(df.isnull(), cbar=False)
```



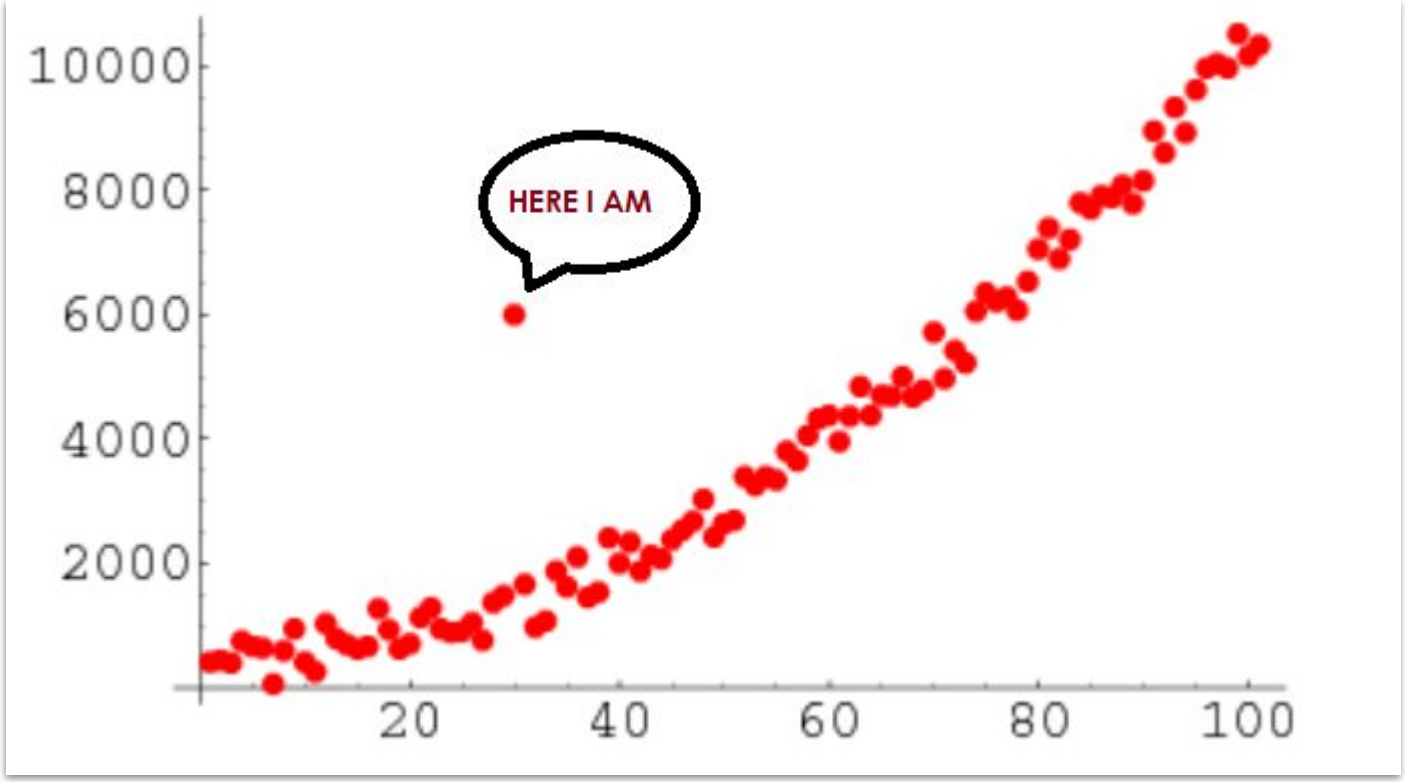
Imputation Strategies

- Deletion
- Mean Imputation
- Median Imputation
- Mode Imputation

Handling Outliers

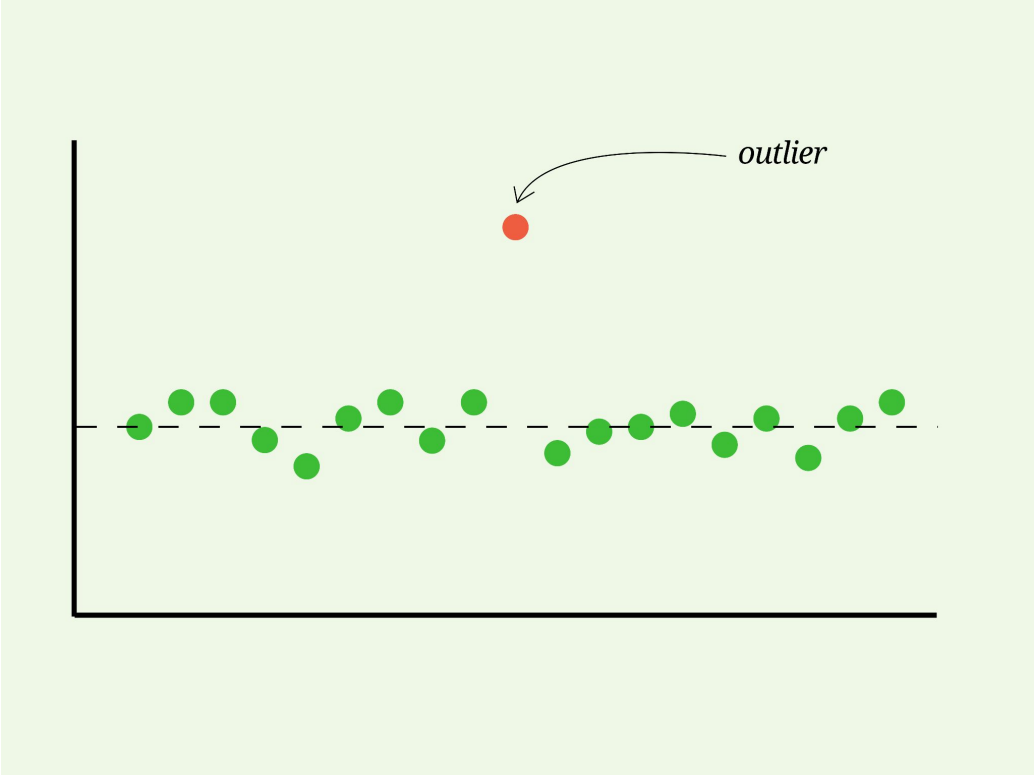
DETECTION & REMOVAL

Outlier



A point that significantly deviates from other observations is an outlier.

Outlier



What is the impact of outlier on mean?

Outlier: https://www.wikihow.com/Calculate-Outliers

Z-Score Analysis

- The Z-score tells you how many standard deviations a data point is from the mean.
- If Z > 3 or Z < -3, it is considered an outlier.

$$Z = \frac{X - \mu}{\sigma}$$

IQR (Interquartile Range)

- More robust than Z-score because it uses quartiles, not mean/std dev.
- Any data point outside the bounds is flagged as an outlier.

IQR = Q3 (75th%) - Q1 (25th%)

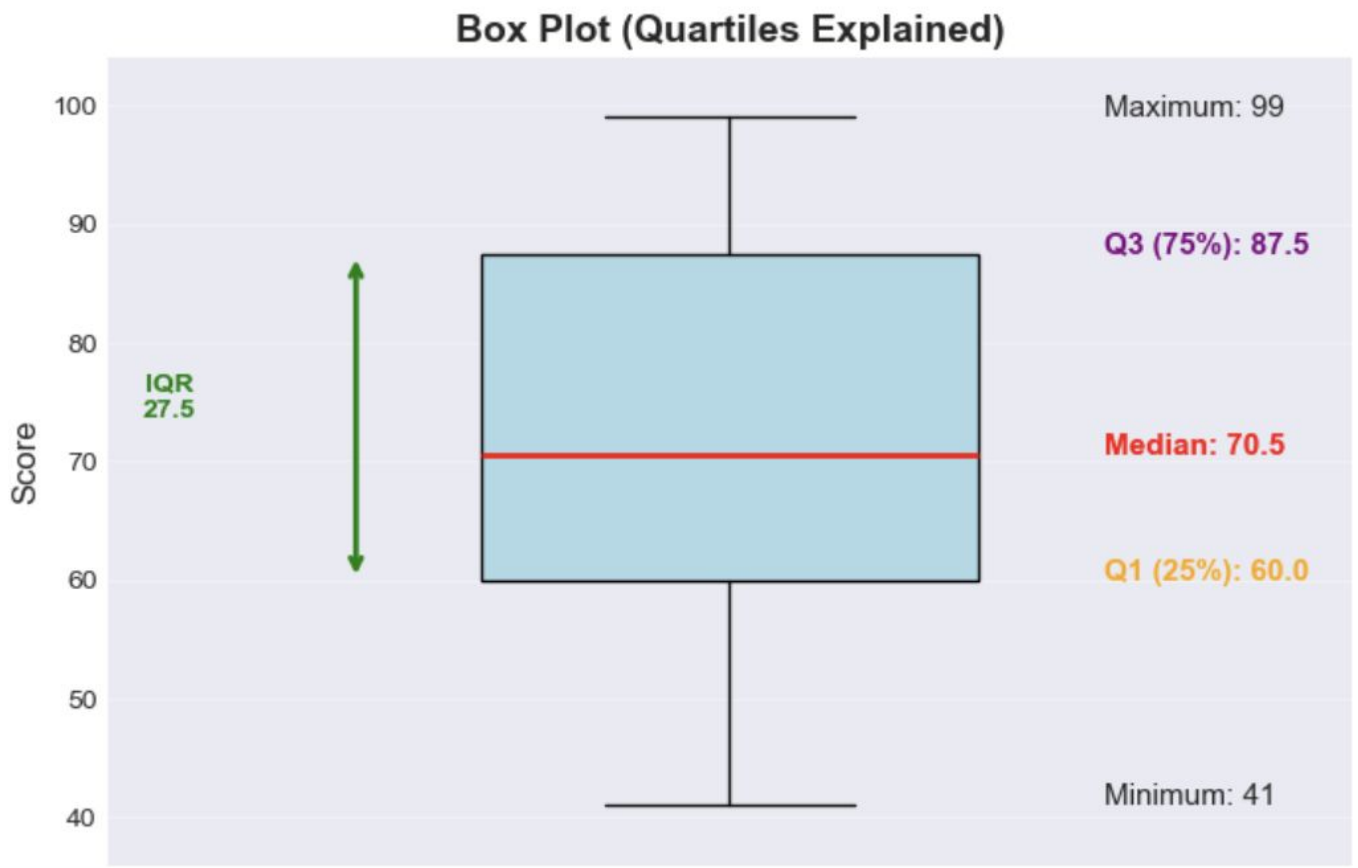
Lower Bound = Q1 - 1.5 * IQR

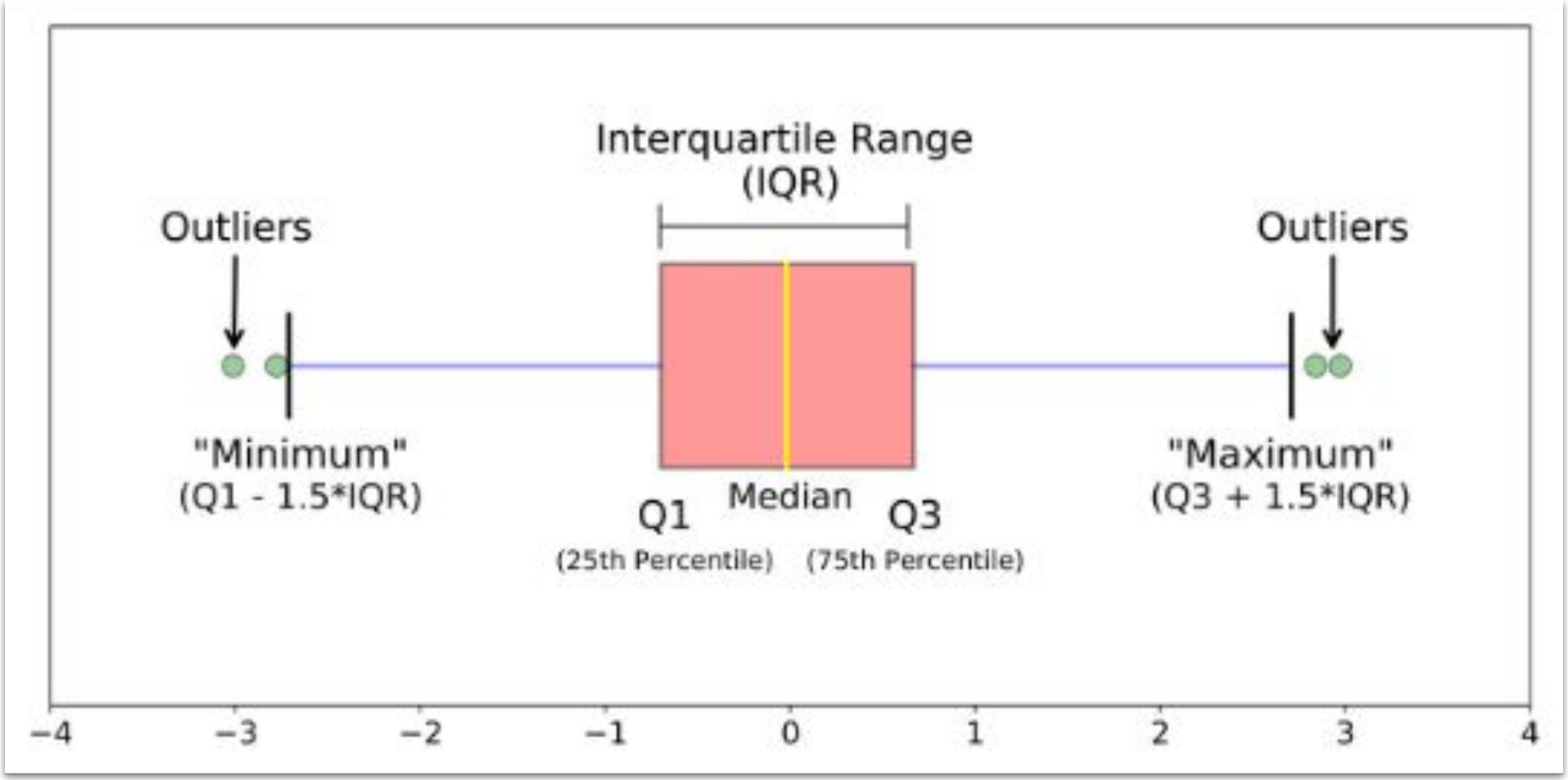
Upper Bound = Q3 + 1.5 * IQR

Visualize IQR

- Quartile Range

  E.g. we have 50 student marks ranging from 40 to 100.




Interquartile Range
(IQR)
Outliers
Outliers
"Minimum"
(Q1 - 1.5*IQR)
Q1
Median
Q3
(25th Percentile)
(75th Percentile)
"Maximum"
(Q3 + 1.5*IQR)
−4
−3
−2
−1
0
1
2
3
4

Handling Strategies

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Trimming

Completely removing the outlier rows.

Pro: Simple.

Con: Data loss.

Capping

Replacing outliers with the upper/lower limit (Winsorizing).

Pro: Preserves data size.

Con: Modifies distribution.

Encoding

CATEGORICAL TO NUMERICAL

Categorical Data Types

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Ordinal Data

Categories with an inherent order.

- Low, Medium, High
- Satisfied, Neutral, Dissatisfied
- Sizes: S, M, L, XL

Nominal Data

Categories with no order.

- Red, Green, Blue
- New York, Paris, Tokyo
- Dog, Cat, Bird

Method 1: Label Encoding

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Use for Ordinal Data

Assigns a unique integer to each category.

Caution: Do NOT use for Nominal data (e.g., Apple=1, Banana=2). The model will think Banana > Apple.

```
from sklearn.preprocessing import
LabelEncoder

le = LabelEncoder()
df['Size_Encoded'] =
le.fit_transform(df['Size'])

# S=0, M=1, L=2
```

Method 2: One-Hot Encoding

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Use for Nominal Data

Creates a binary column for each category.

drop_first=True prevents multicollinearity (Dummy Variable Trap).

```
df = pd.get_dummies(df,
  columns=['Color'],
  drop_first=True)
```

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Q&A

Thank You!